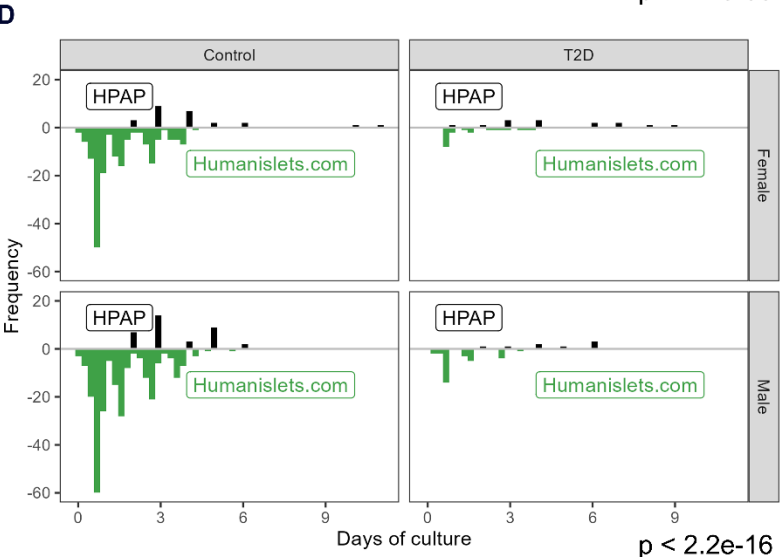
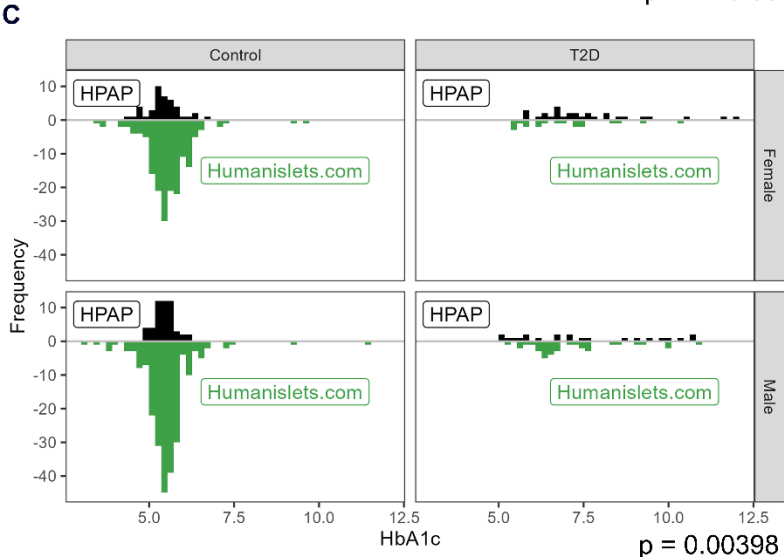
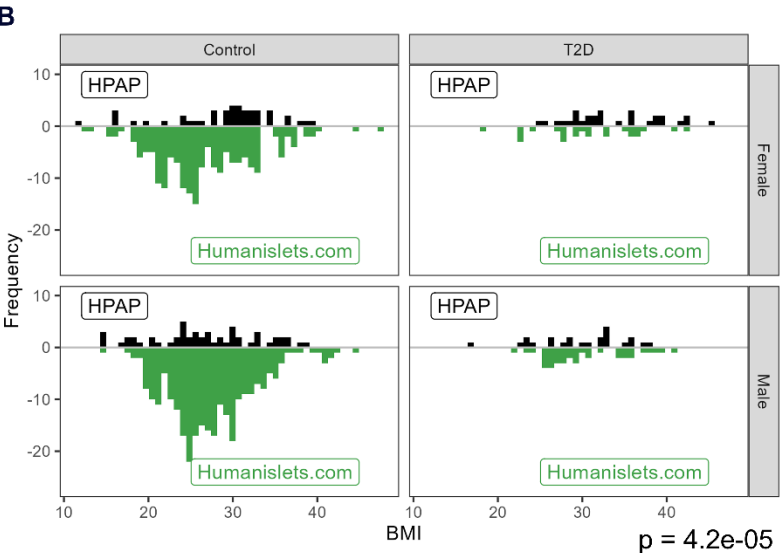
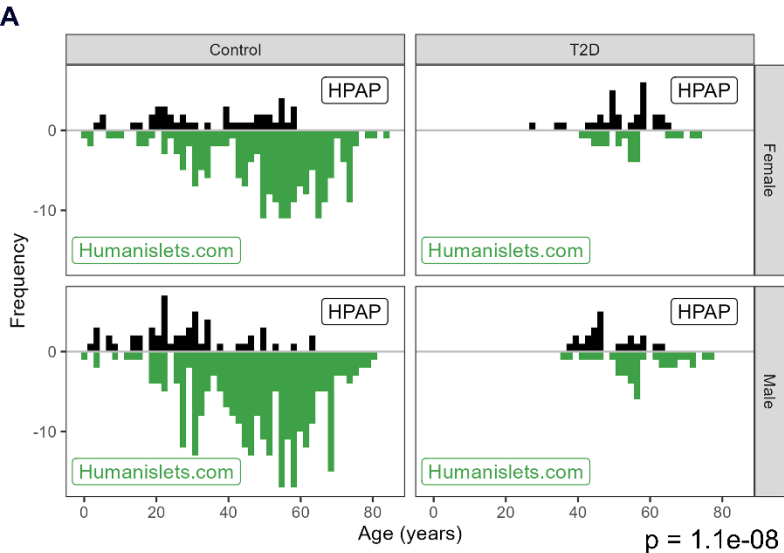


1 Supplemental Figures

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4 **Figure S1. Metadata distributions in HPAP and Humanislets.com datasets.**

5 Distribution of age (A), BMI (B), HbA1c (C), and culture duration (D). Culture duration was not available for all donors and only
6 available data are shown. Distributions were compared by Kolgomorov-Smirnov test and p -values are shown in figure.

7

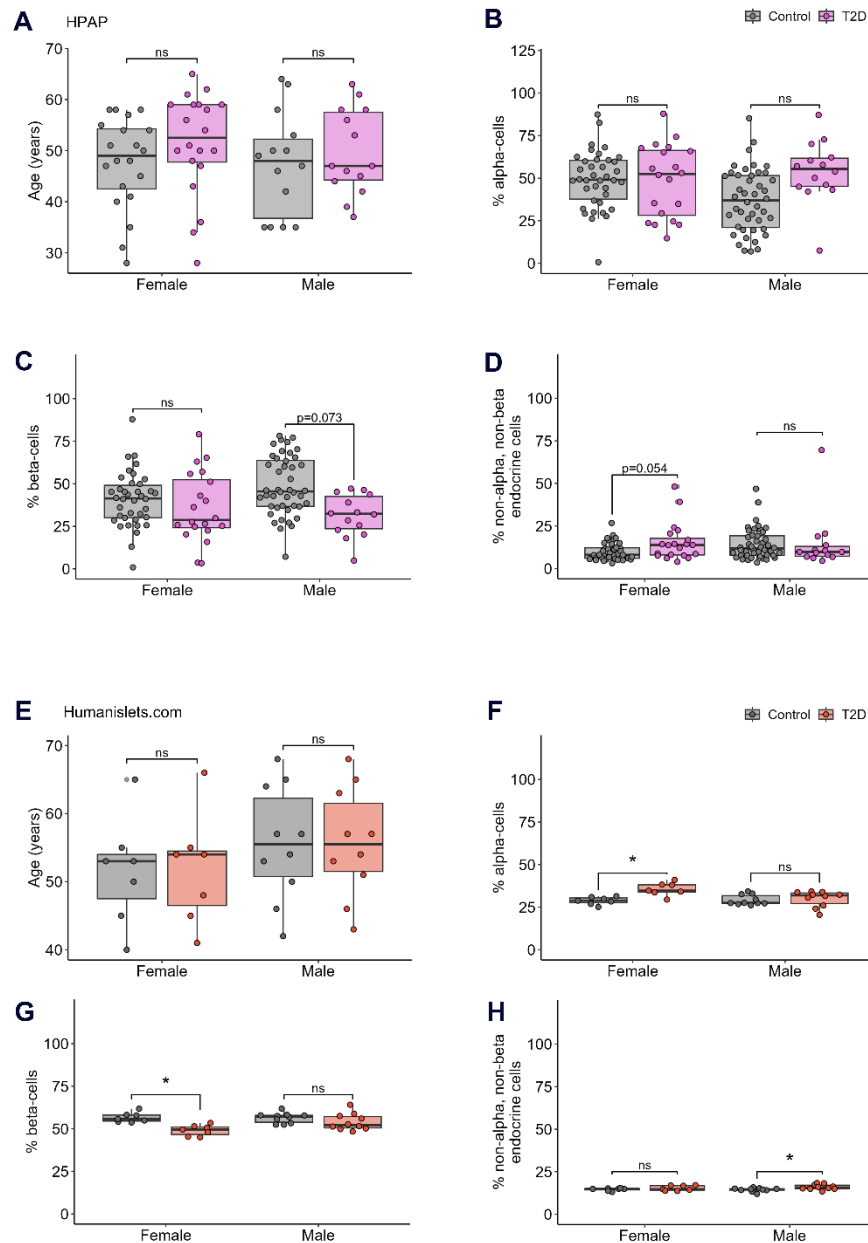


Figure S2. Islet endocrine cell proportions in age-matched data.

Donors who lived with T2D were age-matched with donors without T2D within each sex.

Proportions were determined by CyToF in the HPAP dataset or deconvolution from whole islet proteomics in the Humanislets.com dataset. Donor ages in age-matched dataset (A), alpha-cell proportion (B), beta-cell proportion (C), and non-alpha, non-beta cell proportion (D) in age-

matched HPAP data. Donor ages in age-matched dataset (E), alpha-cell proportion (F), beta-cell

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- 15 proportion (G), and non-alpha, non-beta cell proportion (H) in age-matched Humanislets.com
16 data. * indicates $p < 0.05$, ns = not significant, $p \geq 0.05$.

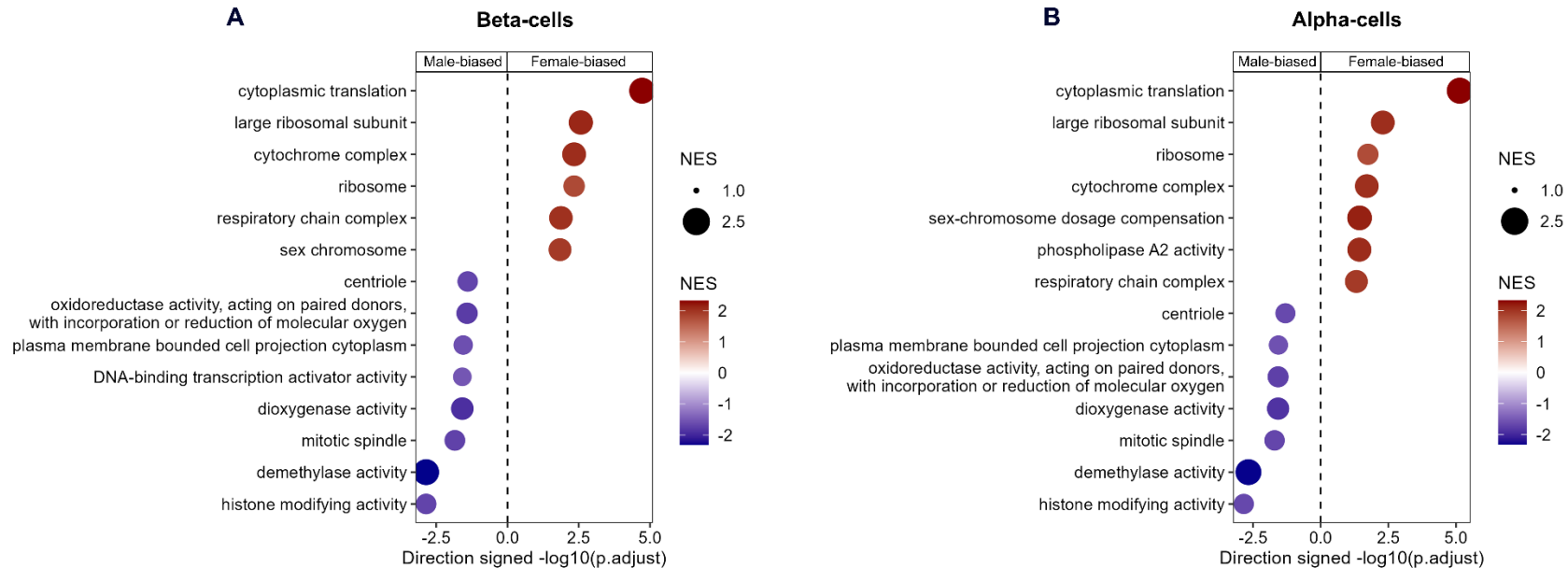
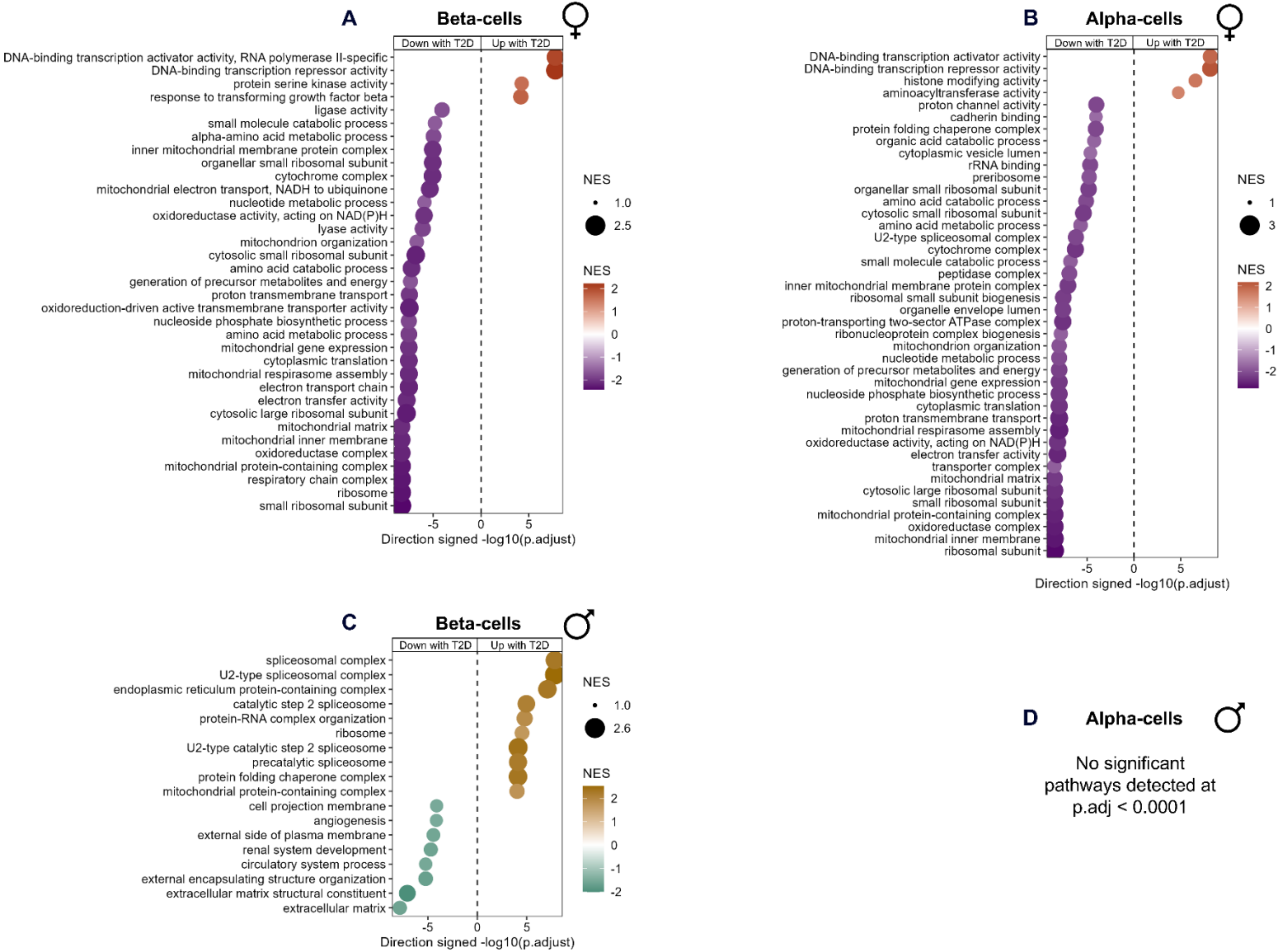


Figure S3. Pseudobulk beta-cell and alpha-cell gene expression in islets from female versus male donors without diabetes aged 15-39.

Significantly altered pathways according to sex in GSEA analysis in HPAP pseudobulk beta-cell scRNAseq data (A) and HPAP pseudobulk alpha-cell scRNAseq data (B). n = 7 female, 17 male. NES = normalized enrichment score.



25 **Figure S4. Pseudobulk beta-cell and alpha-cell gene expression in islets from donors without diabetes versus donors who**
26 **lived with T2D of all ages.**
27 Significantly altered pathways according to T2D status in GSEA analysis at $p < 0.0001$ in HPAP pseudobulk beta-cell scRNAseq data
28 (A) and HPAP pseudobulk alpha-cell scRNAseq data (B) in females. Significantly altered pathways according to T2D status in GSEA
29 analysis at $p < 0.0001$ in HPAP pseudobulk beta-cell scRNAseq data in males (C). No significant pathways at $p < 0.0001$ were detected
30 in HPAP pseudobulk alpha-cell scRNAseq data in males (D). $n = 16$ female control, 11 female T2D, 24 male control, and 7 male T2D
31 donors. NES = normalized enrichment score.

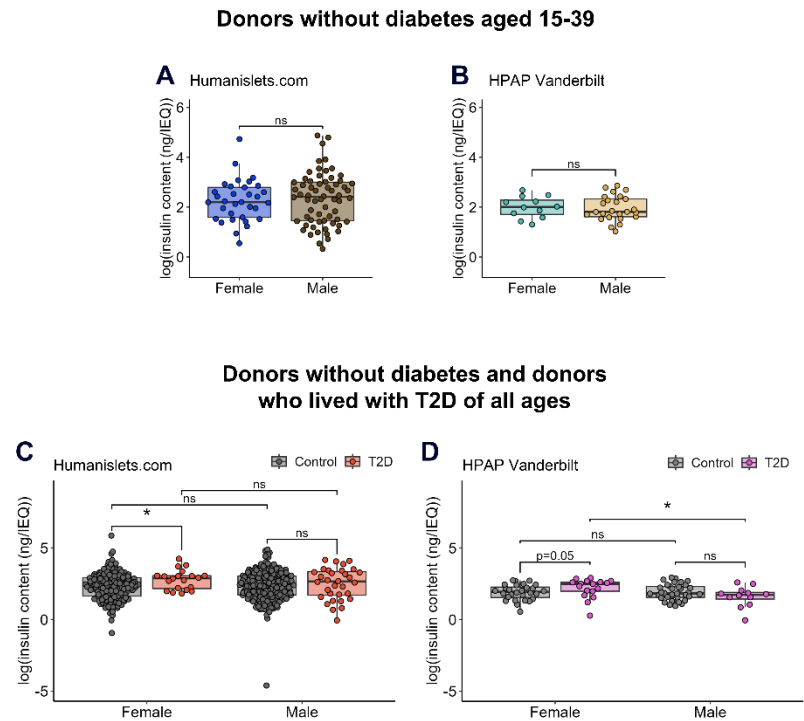
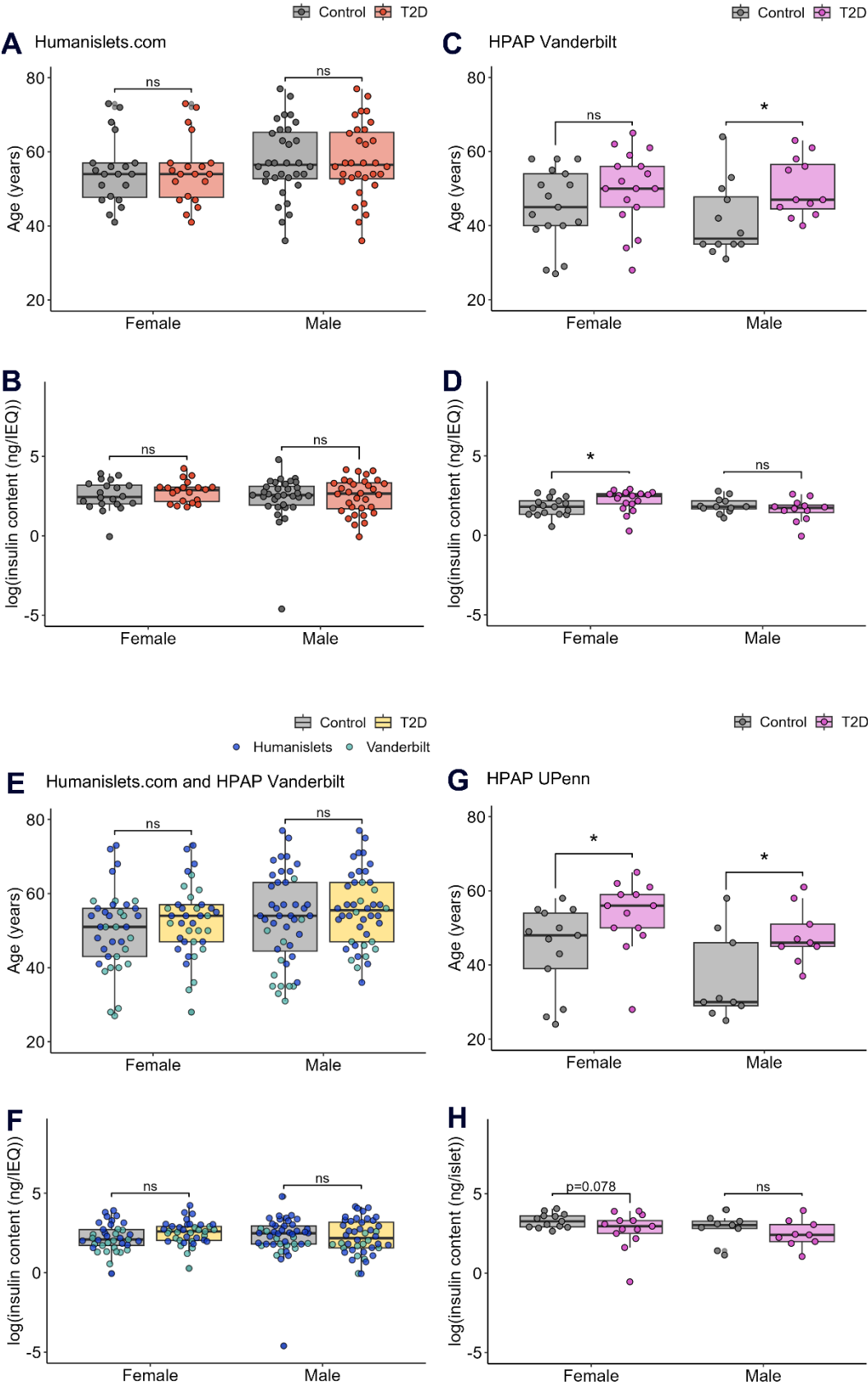


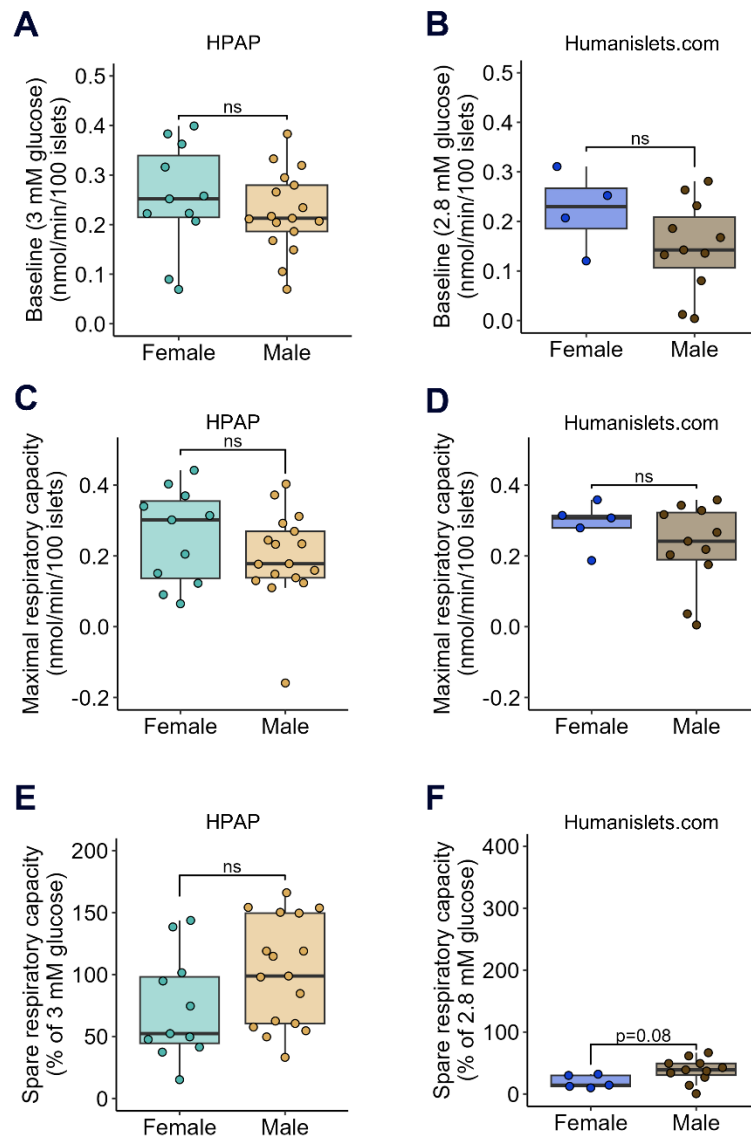
Figure S5. Individual dataset analysis for islet insulin content.

Insulin content from Humanislets.com (A) and HPAP Vanderbilt (B) datasets analyzed individually for donors without diabetes aged 15-39. Insulin content from Humanislets.com (C) and HPAP Vanderbilt (D) datasets analyzed individually for donors without diabetes and donors with T2D of all ages. * indicates $p < 0.05$, ns = not significant, $p \geq 0.05$.



41 **Figure S6. Islet insulin content in age-matched data.**

42 Donors who lived with T2D were age-matched as closely as possible with donors without T2D
43 within each sex. Donor ages in age-matched data (A) and insulin content (B) in age-matched
44 Humanislets.com data. Donor ages in age-matched data (C) and insulin content (D) in age-
45 matched HPAP Vanderbilt data. Donor ages in age-matched data (E) and insulin content (F) in
46 age-matched combined Humanislets.com and HPAP Vanderbilt data. Donor ages in age-
47 matched data (G) and insulin content (H) in age-matched HPAP UPenn data. * indicates $p < 0.05$,
48 ns = not significant, $p \geq 0.05$.



49

50 **Figure S7. Individual dataset analysis for oxygen consumption rate.**

51 For islets from donors without diabetes aged 15-39, baseline respiration from HPAP (A) and

52 Humanislets.com (B), maximal respiratory capacity from HPAP (C) and Humanislets.com (D),

53 and spare respiratory capacity from HPAP (E) and Humanislets.com (F), with datasets analyzed

54 individually. ns = not significant, $p \geq 0.05$.

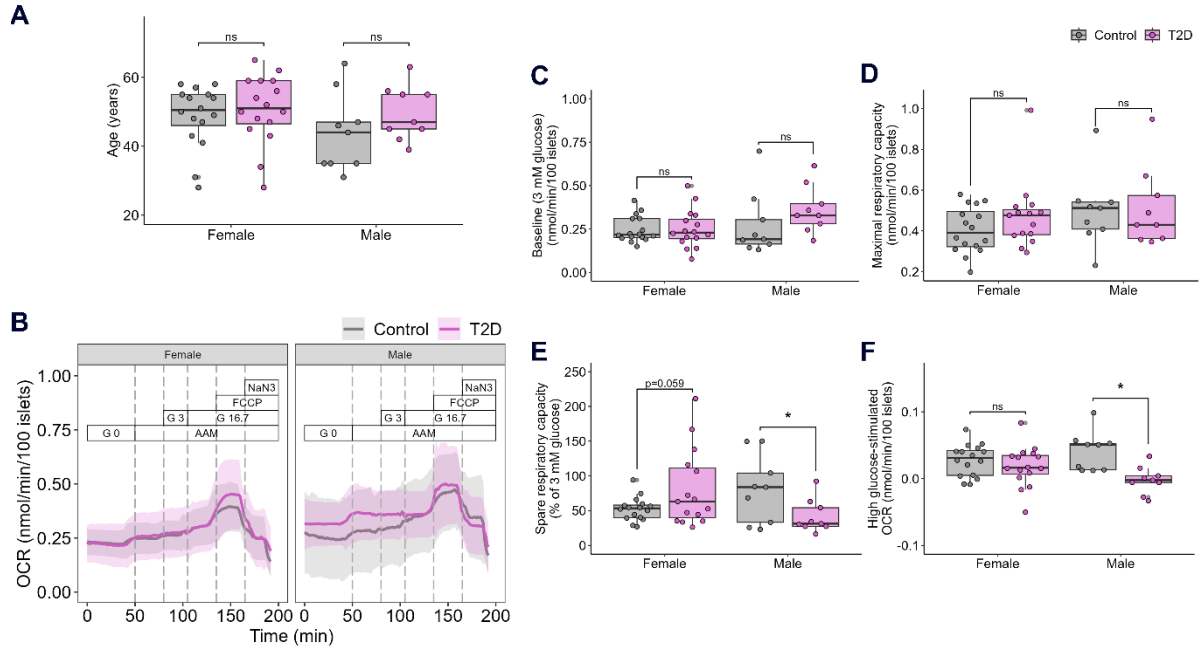


Figure S8. HPAP oxygen consumption rate in age-matched data.

Donors who lived with T2D were age-matched with donors without T2D within each sex. Donor ages in age-matched data (A) and average OCR tracings $\pm$ SD for female and male islets for age-matched data (B). Summary data for baseline OCR (C), maximal respiratory capacity (D), spare respiratory capacity (E), and high glucose-stimulated OCR (F) for age-matched data.

AAM = amino acid mix (0.44 mM alanine, 0.19 mM arginine, 0.038 mM aspartate, 0.094 mM citrulline, 0.12 mM glutamate, 0.30 mM glycine, 0.077 mM histidine, 0.094 mM isoleucine, 0.16 mM leucine, 0.37 mM lysine, 0.05 mM methionine, 0.70 mM ornithine, 0.08 mM phenylalanine, 0.35 mM proline, 0.57 mM serine, 0.27 mM threonine, 0.073 mM tryptophan, and 0.20 mM valine, 2 mM glutamine), G 3 = 3 mM glucose, G 16.7 = 16.7 mM glucose, FCCP = carbonyl cyanide-p-trifluoromethoxyphenylhydrazone (mitochondrial uncoupler), NaN3 = sodium azide (cytochrome oxidase inhibitor), AntA = antimycin A (complex III inhibitor). * indicates $p < 0.05$, ns = not significant, $p \geq 0.05$.

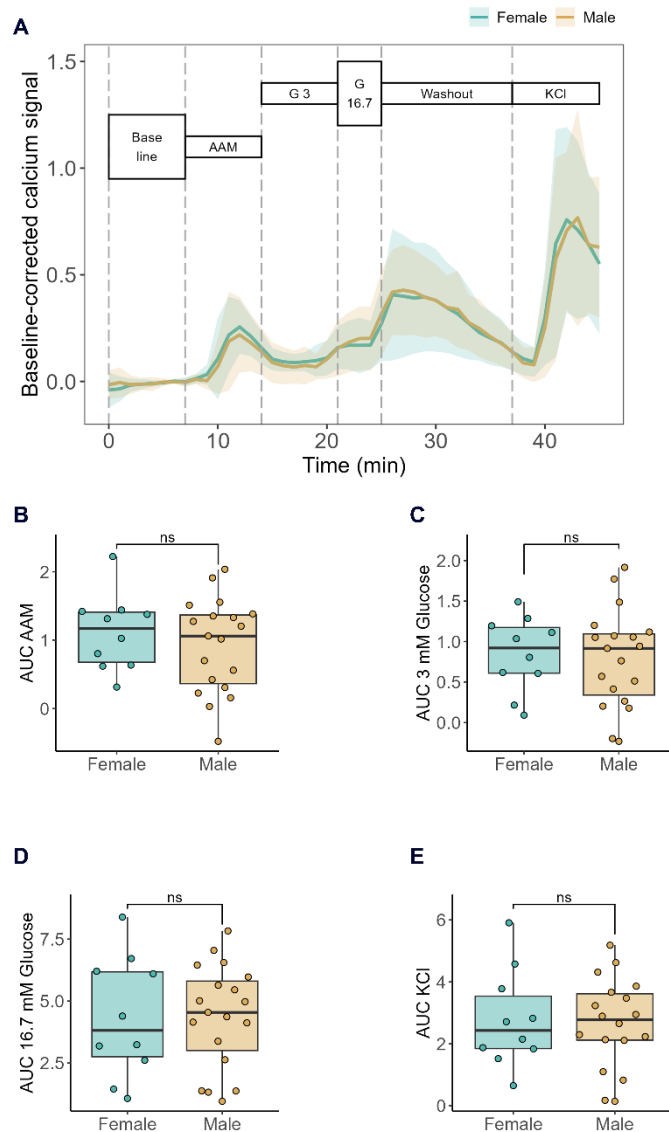


Figure S9. HPAP intracellular calcium imaging data for islets from donors without diabetes aged 15-39.

Average baseline-corrected intracellular calcium signal $\pm$ SD for female and male islets from donors without diabetes aged 15-39 (A). Area under the curve (AUC) data for AAM-stimulated (B), low glucose-stimulated (C), high glucose-stimulated (D), and KCl-stimulated (E) calcium signal. AAM = amino acid mix (0.44 mM alanine, 0.19 mM arginine, 0.038 mM aspartate, 0.094 mM citrulline, 0.12 mM glutamate, 0.30 mM glycine, 0.077 mM histidine, 0.094 mM isoleucine, 0.16 mM leucine, 0.37 mM lysine, 0.05 mM methionine, 0.70 mM ornithine, 0.08 mM

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78 phenylalanine, 0.35 mM proline, 0.57 mM serine, 0.27 mM threonine, 0.073 mM tryptophan, and
79 0.20 mM valine, 2 mM glutamine), G 3 = 3 mM glucose, G 16.7 = 16.7 mM glucose. ns = not
80 significant, $p \geq 0.05$.

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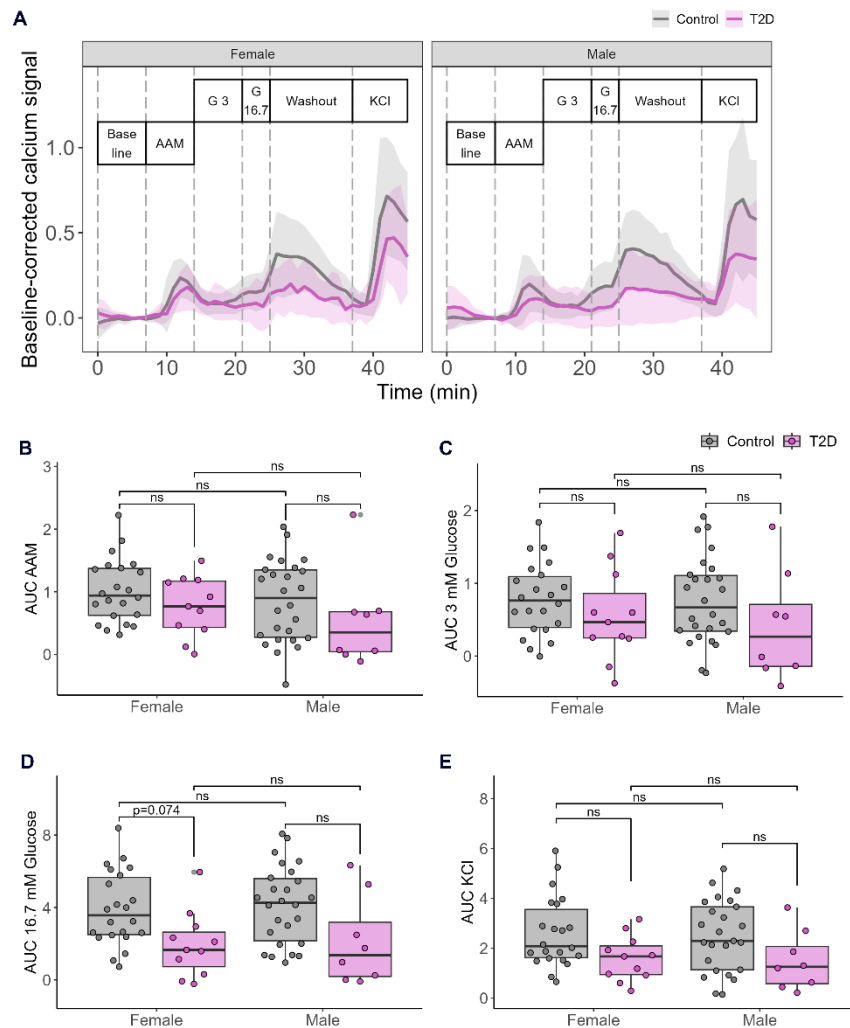


Figure S10. HPAP intracellular calcium imaging data for islets from donors without diabetes and donors with T2D of all ages.

Average baseline-corrected intracellular calcium signal $\pm$ SD for control and T2D, female and male islets (A). Area under the curve (AUC) data for AAM-stimulated (B), low glucose-stimulated (C), high glucose-stimulated (D), and KCl-stimulated (E) calcium signal. AAM = amino acid mix (0.44 mM alanine, 0.19 mM arginine, 0.038 mM aspartate, 0.094 mM citrulline, 0.12 mM glutamate, 0.30 mM glycine, 0.077 mM histidine, 0.094 mM isoleucine, 0.16 mM leucine, 0.37 mM lysine, 0.05 mM methionine, 0.70 mM ornithine, 0.08 mM phenylalanine, 0.35 mM proline,

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- 91 0.57 mM serine, 0.27 mM threonine, 0.073 mM tryptophan, and 0.20 mM valine, 2 mM
92 glutamine), G 3 = 3 mM glucose, G 16.7 = 16.7 mM glucose. ns = not significant, $p \geq 0.05$.

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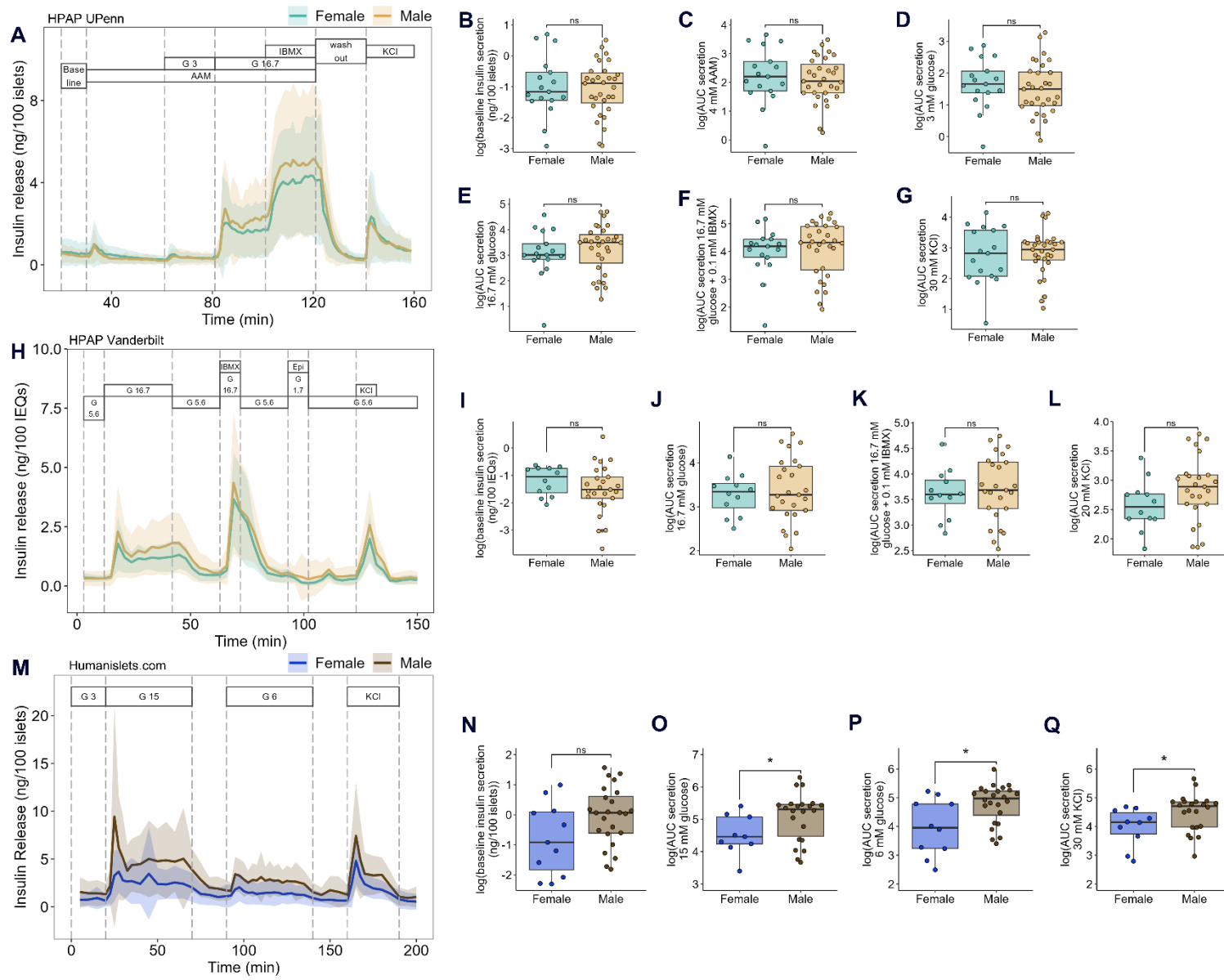


Figure S11. HPAP and Humanislets.com dynamic insulin secretion data following glucose stimulation for islets from donors without diabetes aged 15-39.

Perifusion curves (A), area under the curve for baseline (B), amino acid-induced (C), low glucose-induced (D), high glucose-induced (E), IBMX-induced (F), and KCl-induced (G) insulin secretion for HPAP UPenn data. Perifusion curves (H), baseline insulin secretion (I), area under the curve for high glucose-induced (J), IBMX-induced (K) and KCl-induced (L) insulin secretion for HPAP Vanderbilt data. Perifusion curves (M), baseline insulin secretion (N), area under the curve for high glucose-induced (O), low glucose induced (P), and KCl-induced (Q) insulin secretion for Humanislets.com data. AAM = amino acid mix (0.44 mM alanine, 0.19 mM arginine, 0.038 mM aspartate, 0.094 mM citrulline, 0.12 mM glutamate, 0.30 mM glycine, 0.077 mM histidine, 0.094 mM isoleucine, 0.16 mM leucine, 0.37 mM lysine, 0.05 mM methionine, 0.70 mM ornithine, 0.08 mM phenylalanine, 0.35 mM proline, 0.57 mM serine, 0.27 mM threonine, 0.073 mM tryptophan, and 0.20 mM valine, 2 mM glutamine), G = glucose, IBMX = 3-isobutyl-1-methylxanthine at 0.1 mM. KCl is at 30 mM for UPenn and Humanislets.com, 20 mM for Vanderbilt. Perifusion curves show mean $\pm$ SD. * indicates $p < 0.05$, ns = not significant, $p \geq 0.05$.

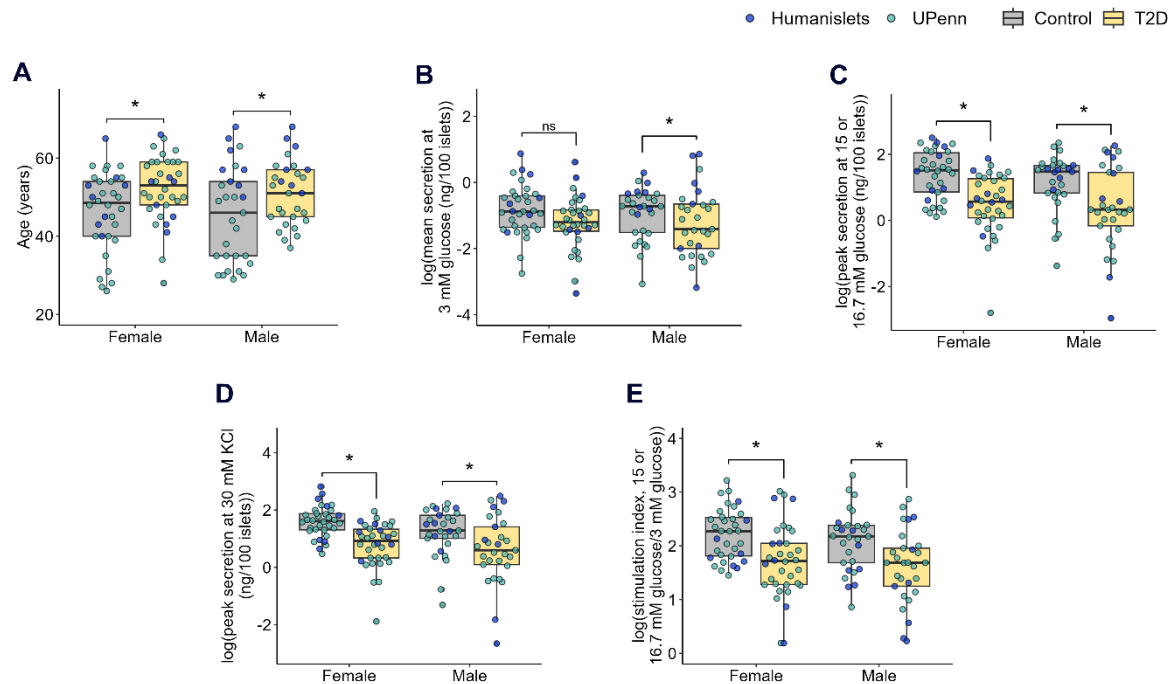


Figure S12. Combined HPAP UPenn and Humanislets.com dynamic insulin secretion data in age-matched data.

Donors who lived with T2D were age-matched with donors without T2D within each sex and dataset. Donor ages in age-matched data (A). Average insulin secretion at 3 mM glucose for age-matched combined HPAP UPenn (AAM also present) and Humanislets.com data (C). Peak insulin secretion at high glucose for age-matched combined HPAP UPenn (16.7 mM glucose plus AAM) and Humanislets.com data (15 mM glucose) (D). Peak insulin secretion at 30 mM KCl for age-matched combined HPAP UPenn and Humanislets.com data (E). Stimulation index calculated as peak high glucose-stimulated secretion divided by average secretion at 3 mM glucose $\pm$ AAM for age-matched combined HPAP UPenn and Humanislets.com data (F). * indicates $p < 0.05$, ns = not significant, $p \geq 0.05$.

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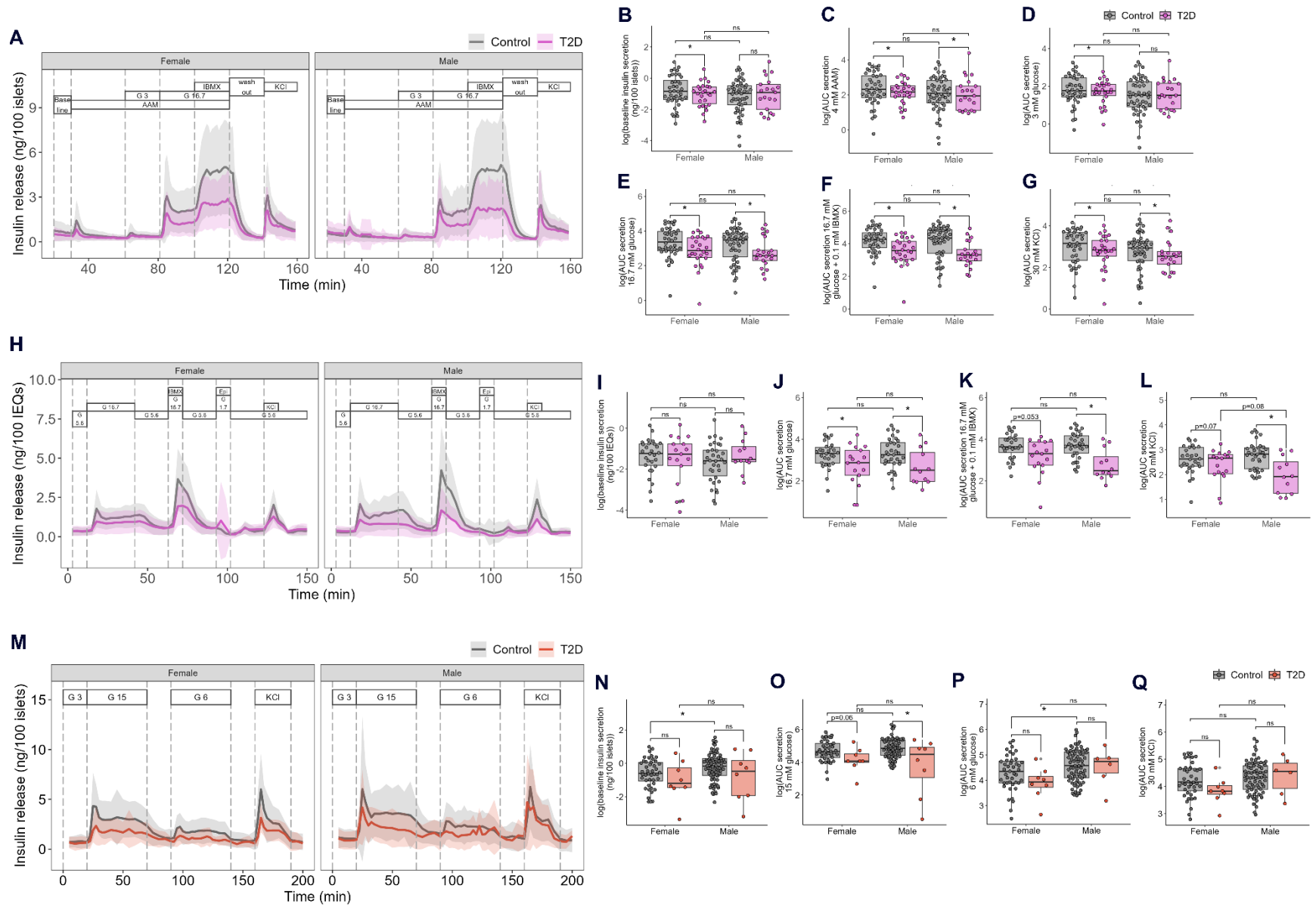


Figure S13. Individual dataset analysis for HPAP UPenn, HPAP Vanderbilt, and Humanislets.com dynamic insulin secretion data for islets from donors without diabetes and donors with T2D of all ages.

Perifusion curves (A), average baseline (0 mM glucose, B), and area under the curve for amino acid-induced (C), 3 mM glucose-induced (D), 16.7 mM glucose-induced (E), 16.7 mM glucose and IBMX-induced (F), and 30 mM KCl-induced (G) insulin secretion for HPAP UPenn data. Perifusion curves (H), average baseline (5.6 mM glucose, I), and area under the curve for 16.7 mM glucose-induced (J), 16.7 mM glucose and IBMX-induced (K) and 20 mM KCl-induced (L) insulin secretion for HPAP Vanderbilt data. Perifusion curves (M), average baseline (3 mM glucose, N), and area under the curve for 15 mM glucose-induced (O), 6 mM glucose-induced (P) and 30 mM KCl-induced (Q) insulin secretion for Humanislets.com data. AAM = amino acid mix (0.44 mM alanine, 0.19 mM arginine, 0.038 mM aspartate, 0.094 mM citrulline, 0.12 mM glutamate, 0.30 mM glycine, 0.077 mM histidine, 0.094 mM isoleucine, 0.16 mM leucine, 0.37 mM lysine, 0.05 mM methionine, 0.70 mM ornithine, 0.08 mM phenylalanine, 0.35 mM proline, 0.57 mM serine, 0.27 mM threonine, 0.073 mM tryptophan, and 0.20 mM valine, 2 mM glutamine), G = glucose, IBMX = 3-isobutyl-1-methylxanthine at 0.1 mM. Perifusion curves show mean $\pm$ SD. * indicates $p < 0.05$, ns = not significant, $p \geq 0.05$.

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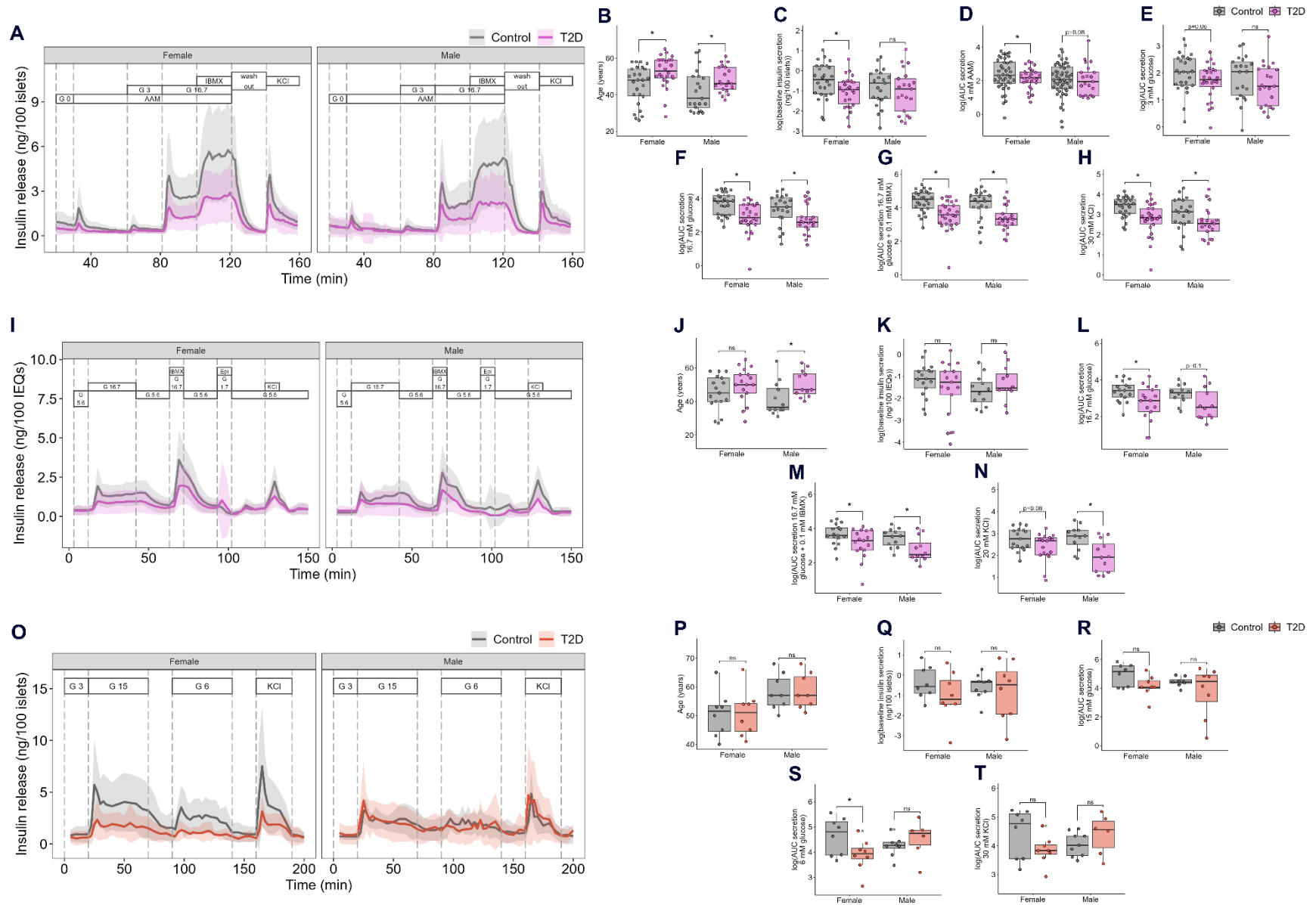


Figure S14. Individual dataset analysis for HPAP UPenn, HPAP Vanderbilt, and Humanislets.com dynamic insulin secretion in age-matched data.

Donors who lived with T2D were age-matched as closely as possible with donors without T2D within each sex. Perfusion curves (A), donor ages (B), average baseline (0 mM glucose, C), and area under the curve for amino acid-induced (D), 3 mM glucose-induced (E), 16.7 mM glucose-induced (F), 16.7 mM glucose and IBMX-induced (G), and 30 mM KCl-induced (H) insulin secretion for age-matched HPAP UPenn data. Perfusion curves (I), donor ages (J), average baseline (5.6 mM glucose, K), and area under the curve for 16.7 mM glucose-induced (L), 16.7 mM glucose and IBMX-induced (M) and 20 mM KCl-induced (N) insulin secretion for age-matched HPAP Vanderbilt data. Perfusion curves (O), donor ages (P), average baseline (3 mM glucose, Q), and area under the curve for 15 mM glucose-induced (R), 6 mM glucose-induced (S) and 30 mM KCl-induced (T) insulin secretion for age-matched Humanislets.com data. AAM = amino acid mix (0.44 mM alanine, 0.19 mM arginine, 0.038 mM aspartate, 0.094 mM citrulline, 0.12 mM glutamate, 0.30 mM glycine, 0.077 mM histidine, 0.094 mM isoleucine, 0.16 mM leucine, 0.37 mM lysine, 0.05 mM methionine, 0.70 mM ornithine, 0.08 mM phenylalanine, 0.35 mM proline, 0.57 mM serine, 0.27 mM threonine, 0.073 mM tryptophan, and 0.20 mM valine, 2 mM glutamine), G = glucose, IBMX = 3-isobutyl-1-methylxanthine at 0.1 mM. Perfusion curves show mean $\pm$ SD. * indicates $p < 0.05$, ns = not significant, $p \geq 0.05$.

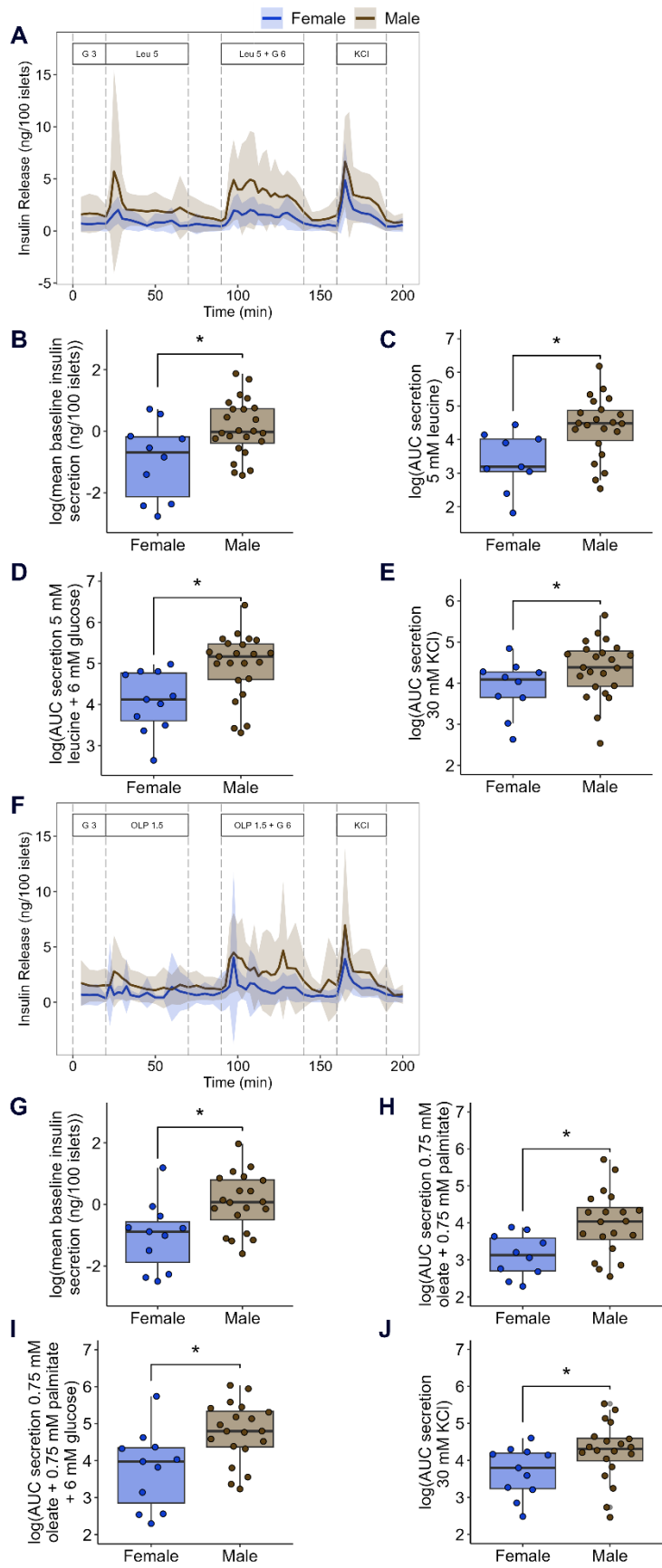
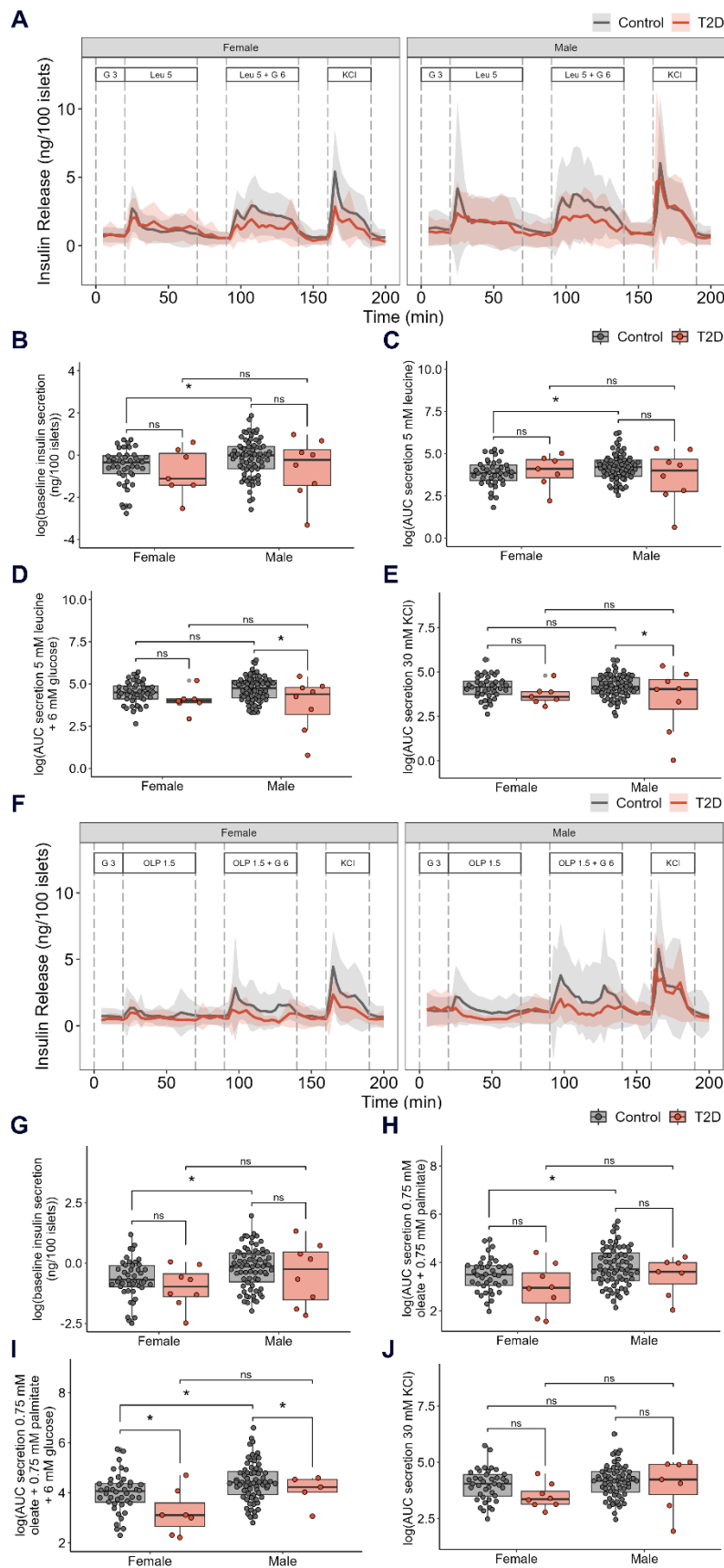


Figure S15. Humanislets.com dynamic insulin secretion data following leucine or fatty acid stimulation for islets from donors without diabetes aged 15-39.

Perifusion curves (A) and summary statistics for baseline insulin secretion (B), leucine-induced insulin secretion (C), nutrient combination-induced insulin secretion (D), and KCl-induced insulin secretion (E) for leucine perifusion. Perifusion curves (F) and summary statistics for baseline insulin secretion (G), oleate/palmitate-induced insulin secretion (H), nutrient combination-induced insulin secretion (I), and KCl-induced insulin secretion (J) for oleate + palmitate perifusion. Perifusion curves show mean $\pm$ SD. * indicates $p < 0.05$, ns = not significant, $p \geq 0.05$.



161 **Figure S16. Humanislets.com dynamic insulin secretion data following leucine or fatty**
162 **acid stimulation for islets from donors without diabetes and donors with T2D of all ages.**
163 Perfusion curves (A) and summary statistics for baseline insulin secretion (B), leucine-induced
164 insulin secretion (C), nutrient combination-induced insulin secretion (D), and KCl-induced insulin
165 secretion (E) for leucine perfusion. Perfusion curves (F) and summary statistics for baseline
166 insulin secretion (G), oleate/palmitate-induced insulin secretion (H), nutrient combination-
167 induced insulin secretion (I), and KCl-induced insulin secretion (J) for oleate + palmitate
168 perfusion. Perfusion curves show mean $\pm$ SD. * indicates $p < 0.05$, ns = not significant, $p \geq 0.05$.

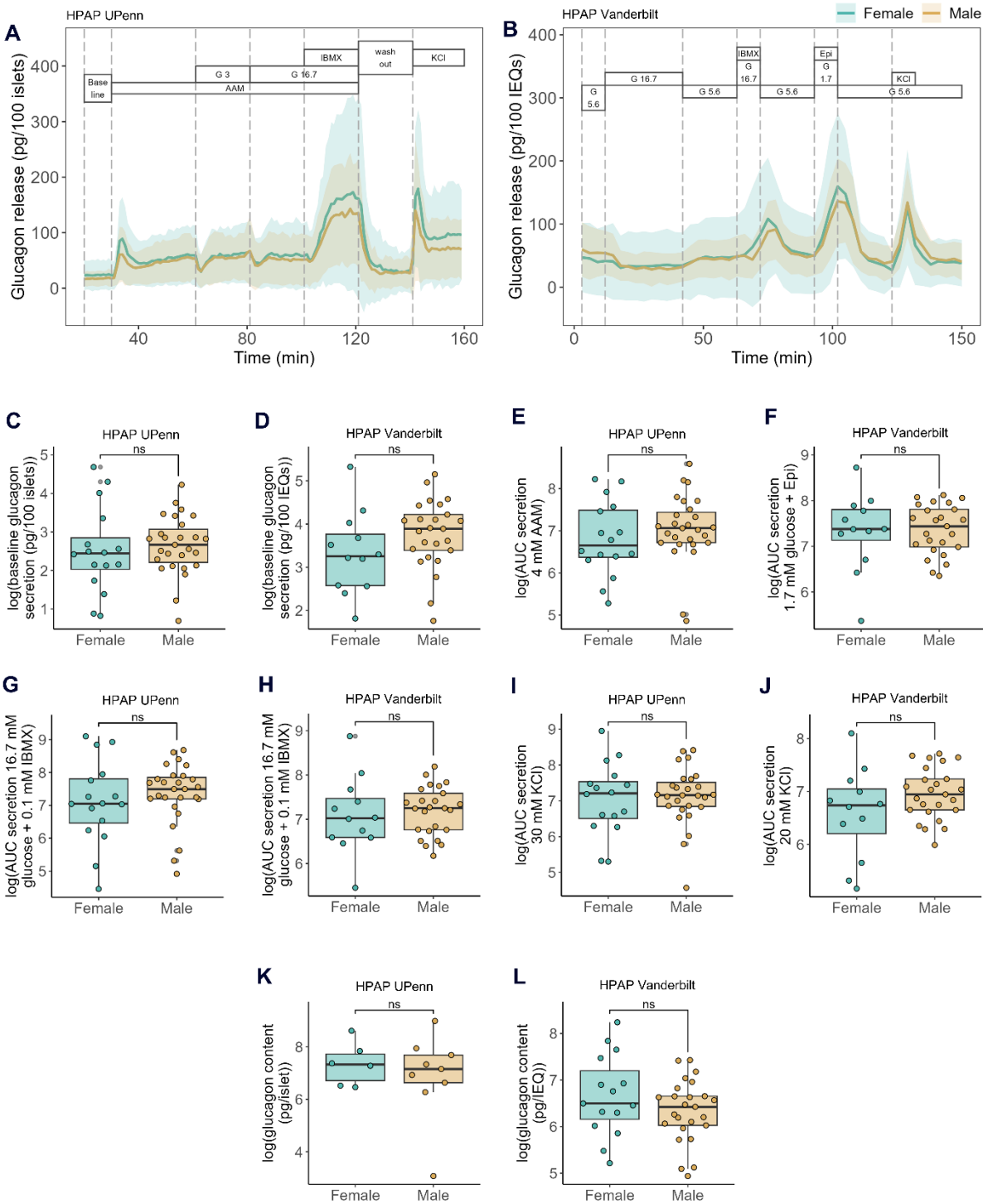


Figure S17. HPAP dynamic glucagon secretion data for islets from donors without diabetes aged 15-39.

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Perifusion curves for HPAP UPenn (A) and HPAP Vanderbilt (B) data. Baseline glucagon secretion from UPenn (C) and Vanderbilt (D), area under the curve for amino acid-stimulated secretion from UPenn (E), low glucose and epinephrine-stimulated secretion from Vanderbilt (F), IBMX-stimulated secretion from UPenn (G) and Vanderbilt (H), and KCl-stimulated secretion from UPenn (I) and Vanderbilt (J). Islet glucagon content from UPenn (K) and Vanderbilt (L). AAM = amino acid mix (0.44 mM alanine, 0.19 mM arginine, 0.038 mM aspartate, 0.094 mM citrulline, 0.12 mM glutamate, 0.30 mM glycine, 0.077 mM histidine, 0.094 mM isoleucine, 0.16 mM leucine, 0.37 mM lysine, 0.05 mM methionine, 0.70 mM ornithine, 0.08 mM phenylalanine, 0.35 mM proline, 0.57 mM serine, 0.27 mM threonine, 0.073 mM tryptophan, and 0.20 mM valine, 2 mM glutamine), G = glucose, IBMX = 3-isobutyl-1-methylxanthine at 0.1 mM. KCl is at 30 mM for UPenn, 20 mM for Vanderbilt. Perifusion curves show mean $\pm$ SD. ns = not significant, $p \geq 0.05$.

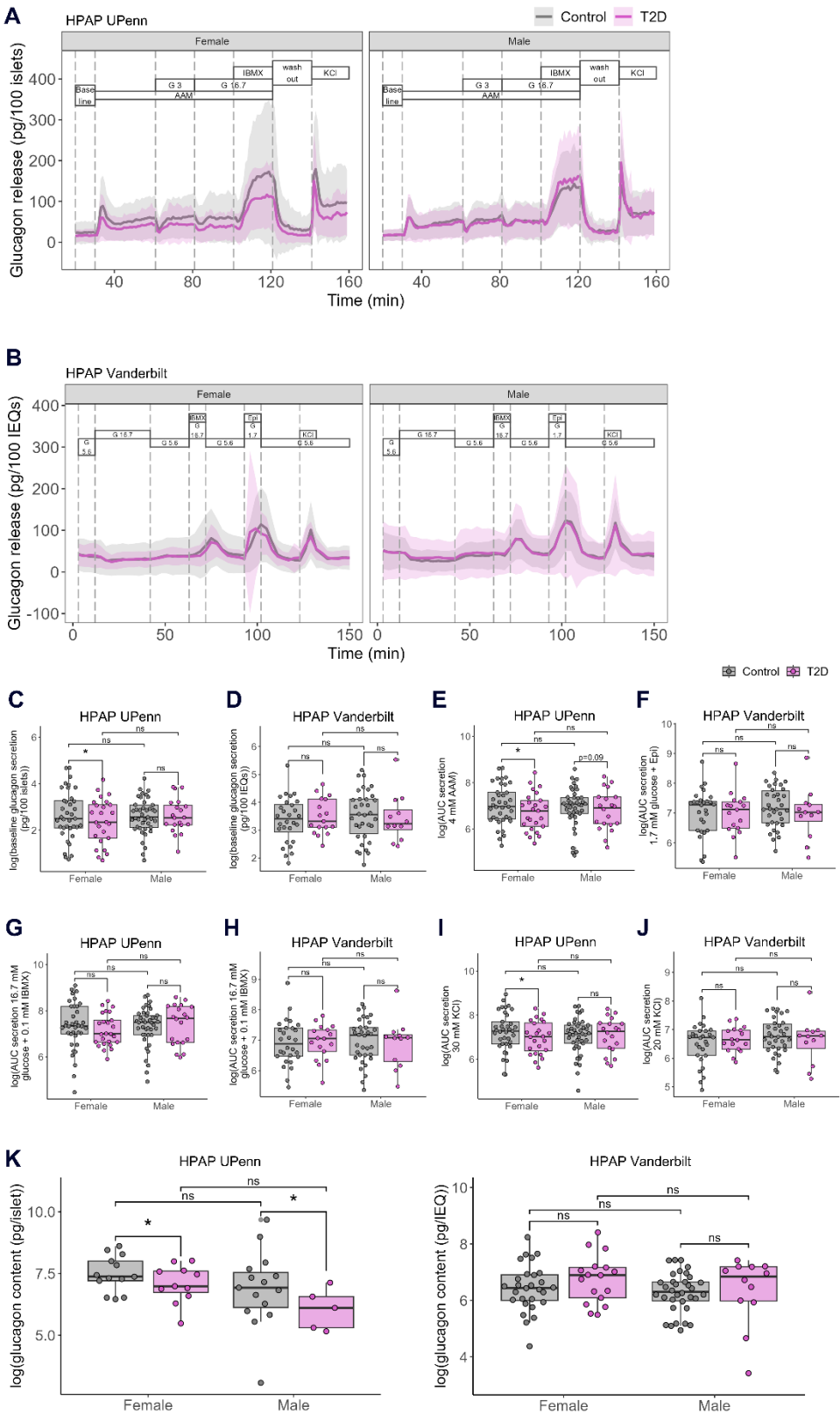


Figure S18. HPAP dynamic glucagon secretion data for islets from donors without diabetes and donors with T2D of all ages.

Perifusion curves for HPAP UPenn (A) and HPAP Vanderbilt (B) data. Baseline glucagon secretion from UPenn (C) and Vanderbilt (D), area under the curve for amino acid-stimulated secretion from UPenn (E), low glucose and epinephrine-stimulated secretion from Vanderbilt (F), IBMX-stimulated secretion from UPenn (G) and Vanderbilt (H), and KCl-stimulated secretion from UPenn (I) and Vanderbilt (J). Islet glucagon content from UPenn (K) and Vanderbilt (L). AAM = amino acid mix (0.44 mM alanine, 0.19 mM arginine, 0.038 mM aspartate, 0.094 mM citrulline, 0.12 mM glutamate, 0.30 mM glycine, 0.077 mM histidine, 0.094 mM isoleucine, 0.16 mM leucine, 0.37 mM lysine, 0.05 mM methionine, 0.70 mM ornithine, 0.08 mM phenylalanine, 0.35 mM proline, 0.57 mM serine, 0.27 mM threonine, 0.073 mM tryptophan, and 0.20 mM valine, 2 mM glutamine), G = glucose, IBMX = 3-isobutyl-1-methylxanthine at 0.1 mM. KCl is at 30 mM for UPenn, 20 mM for Vanderbilt. * indicates $p < 0.05$, ns = not significant, $p \geq 0.05$.

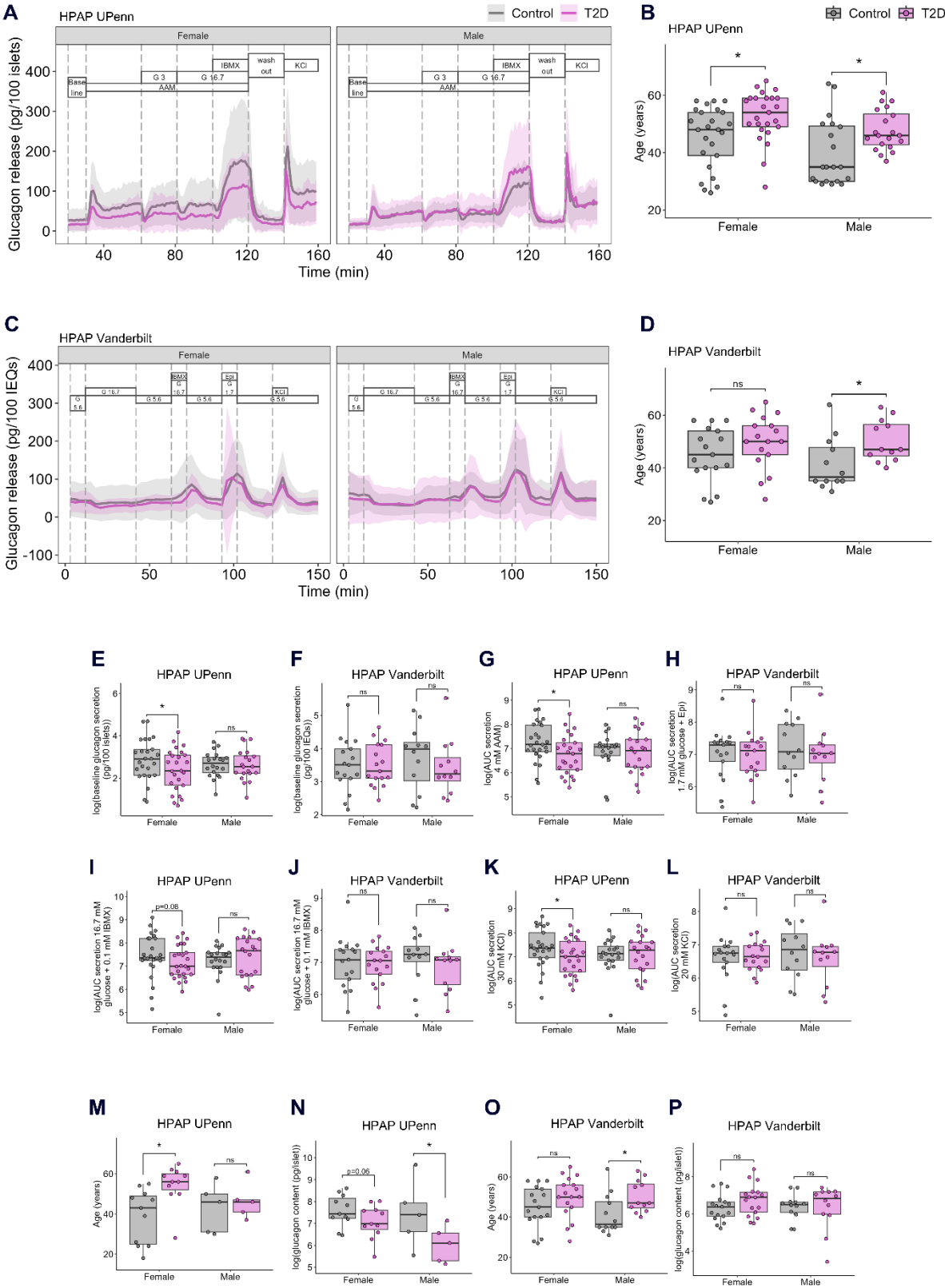


Figure S19. HPAP dynamic glucagon secretion data in age-matched data.

Donors who lived with T2D were age-matched as closely as possible with donors without T2D within each sex. Perifusion curves (A) and donor ages (B) for age-matched HPAP UPenn data. Perifusion curves (C) and donors ages (D) for age-matched HPAP Vanderbilt data. Baseline glucagon secretion from UPenn (E) and Vanderbilt (F), area under the curve for amino acid-stimulated secretion from UPenn (G), low glucose and epinephrine-stimulated secretion from Vanderbilt (C), IBMX-stimulated secretion from UPenn (I) and Vanderbilt (J), and KCl-stimulated secretion from UPenn (K) and Vanderbilt (L) in age-matched data. Donor ages (M) and islet glucagon content (N) for age-matched UPenn data. Donor ages (O) and islet glucagon content (P) for age-matched Vanderbilt data. AAM = amino acid mix (0.44 mM alanine, 0.19 mM arginine, 0.038 mM aspartate, 0.094 mM citrulline, 0.12 mM glutamate, 0.30 mM glycine, 0.077 mM histidine, 0.094 mM isoleucine, 0.16 mM leucine, 0.37 mM lysine, 0.05 mM methionine, 0.70 mM ornithine, 0.08 mM phenylalanine, 0.35 mM proline, 0.57 mM serine, 0.27 mM threonine, 0.073 mM tryptophan, and 0.20 mM valine, 2 mM glutamine), G = glucose, IBMX = 3-isobutyl-1-methylxanthine at 0.1 mM. KCl is at 30 mM for UPenn, 20 mM for Vanderbilt. * indicates $p < 0.05$, ns = not significant, $p \geq 0.05$.

Supplemental Table Legends

Table S1. Full differential expression results for HPAP and Humanislets.com transcriptomics comparing female and male donors without diabetes aged 15-39.

Table S2. Full GSEA results for HPAP and Humanislets.com transcriptomics comparing female and male donors without diabetes aged 15-39.

Table S3. Full differential expression results for Humanislets.com proteomics comparing female and male donors without diabetes aged 15-39.

Table S4. Full GSEA results for Humanislets.com proteomics comparing female and male donors without diabetes aged 15-39.

Table S5. Full differential expression results for HPAP and Humanislets.com transcriptomics comparing female non-diabetic and T2D donors.

Table S6. Full differential expression results for HPAP and Humanislets.com transcriptomics comparing male non-diabetic and T2D donors.

Table S7. Full GSEA results for HPAP and Humanislets.com transcriptomics analyses comparing female non-diabetic and T2D donors.

Table S8. Full GSEA results for HPAP and Humanislets.com transcriptomics comparing male non-diabetic and T2D donors.

Table S9. Full differential expression for Humanislets.com proteomics comparing female non-diabetic and T2D donors.

Table S10. Full differential expression for Humanislets.com proteomics comparing male non-diabetic and T2D donors.

Table S11. Full GSEA results for Humanislets.com proteomics comparing female non-diabetic and T2D donors.

Table S12. Full GSEA results for Humanislets.com proteomics comparing male non-diabetic and T2D donors.